



2025

K24

BOOKLET NO.

103108

Forest Services
Mechanical Engineering

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

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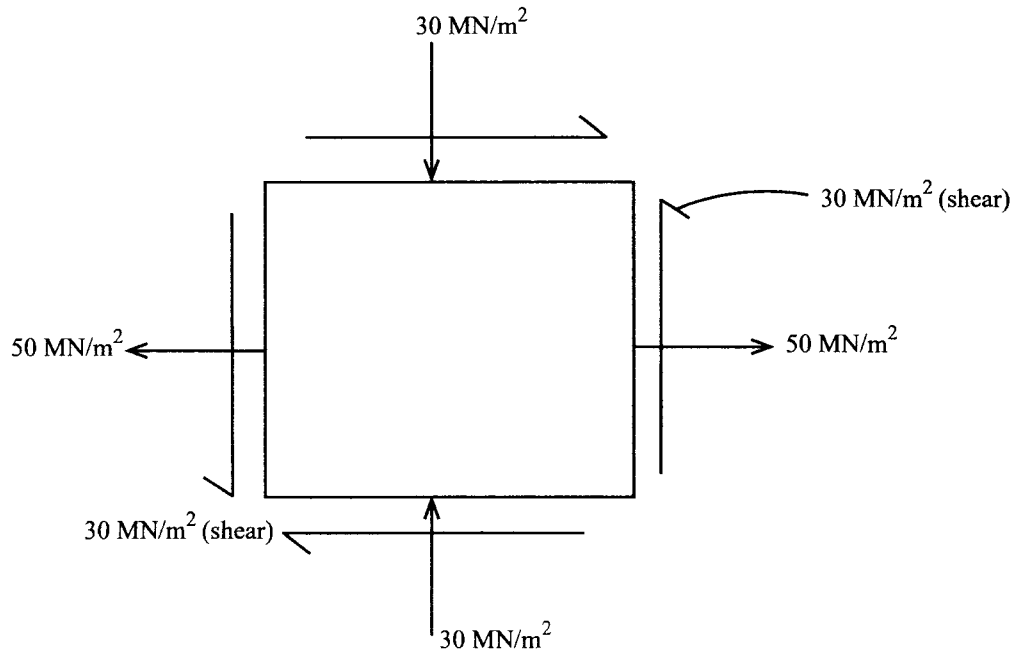


SECTION – A

Q1. Solve any five out of seven.

(8×5=40)

- Q1. (a) The state of stress at a point of a machine is as shown in fig. Determine the principal stresses, maximum shear stress and its inclination. 8



- Q1. (b) Discuss the Shape Memory Alloys (SMA) and their applications. 8

- Q1. (c) Explain with neat sketches the sheet metal operations such as bending, drawing and shearing. 8

- Q1. (d) What is human machine interface and explain its role in SCADA. 8

- Q1. (e) Explain the use of CPM and PERT in project management and key components of a network diagram. 8

- Q1. (f) What is Lagrange interpolation and state the difference between interpolation and extrapolation with examples. 8
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- Q1.** (g) Consider the integral of a simple function, say $f(x) = x^2$ over the interval $[0, 1]$ using the trapezoidal rule. 8

The exact solution can be found as $\int_0^1 x^2 = \frac{1}{3}$.

Answer the following :

- i) What is the significance of uniform portion in numerical integration ?
- ii) Write a pseudocode in 'C' to approximate the given integral using trapezoidal rule.
- iii) In numerical integration, why does increasing the number of sub-intervals improve the accuracy ?

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- Q2.** (a) The centre distance between two spur gears in a mesh is to be approximately 275 mm. The gear ratio is 10 to 1. The smaller gear transmits 360 kW at 1800 rpm. The pressure angle of the involute teeth is 20° and addendum is equal to one module. The limiting value of normal tooth pressure is 1 kN/mm of width. Determine :

- i) The nearest standard module so that interference does not occur.
- ii) Number of teeth on smaller and larger wheel.
- iii) Width of smaller wheel. 15

- Q2.** (b) Explain the concept of a phase diagram. Describe the iron-carbon equilibrium diagram in detail with phases and transformations. 15

- Q2.** (c) Discuss the problems of unbalanced assembly lines and their impact on productivity and efficiency of production systems. 10

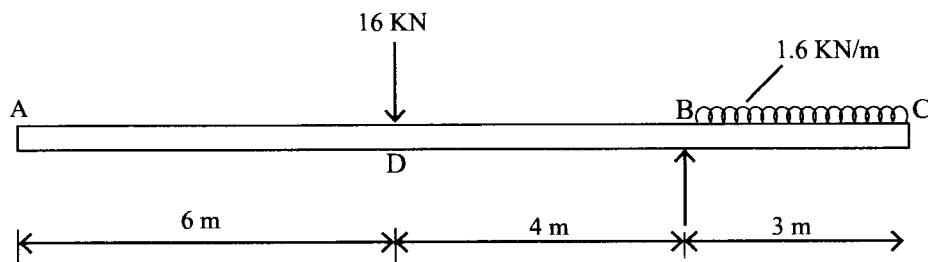


Q3. (a) A beam ABC is loaded as shown in the given figure.

If $E = 200 \times 10^6 \text{ KN/m}^2$ and $I = 9 \times 10^{-5} \text{ m}^4$

Determine :

- i) Slope at the end 'A'.
- ii) Deflection at the free end 'C'.
- iii) The maximum deflection.



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Q3. (b) Define additive manufacturing and explain its principles, features, different types and advantages over subtractive manufacturing processes.

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Q3. (c) Explain the following characteristics of measurements with suitable examples. (Any five)

- i) Accuracy
- ii) Precision
- iii) Linearity
- iv) Sensitivity
- v) Repeatability
- vi) Hysteresis

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Q4. (a) Define reliability engineering. Explain its importance, objectives and applications in engineering systems.

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Q4. (b) What is computer integrated manufacturing? Explain its architecture with a neat block diagram. Also discuss its major components such as CAD, CAM etc.

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Q4. (c) Discuss the principle of ECM with sketch and discuss its advantages, limitations and applications.

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SECTION – B

- Q5. Attempt **any five** from seven. (8×5=40)
- Q5. (a) Comment on irreversibility and Tds relations. 8
- Q5. (b) Establish relation between heat balance, station heat rate and plant heat rate of a power plant. 8
- Q5. (c) Prepare a note on mechanical, thermal and volumetric efficiency of an internal combustion engine. 8
- Q5. (d) Prepare a note on alternative fuel, carburetion and fuel injection of an IC engine emphasizing on trends in development. 8
- Q5. (e) Correlate diffusive and convective mass transfer. 8
- Q5. (f) Discuss the significance of ODP and GWP as primary drivers for the recent technological shifts and refrigerant transitions in the HVAC and R sector. 8
- Q5. (g) Summarise the computational fluid dynamics solution approach in brief. 8

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- Q6. (a) A steam power plant operates between a boiler pressure of 42 bar and a condenser pressure of 0.035 bar. Calculate for these limits the cycle efficiency, the work ratio and the specific steam consumption :
- i) For a Carnot cycle using wet steam ;
- ii) For a Rankine cycle with dry saturated steam at entry to the turbine.

Use following steam properties.

- Boiler pressure ($P_1 = 42$ bar)
 $T_1 = T_{\text{sat}} = 526.4$ K
 $h_f = 1104$ kJ/kg, $h_g = 2798$ kJ/kg
 $s_f = 2.825$ kJ/kg.K, $s_g = 6.049$ kJ/kg
- Condenser pressure ($P_2 = 0.035$ bar)
 $T_2 = T_{\text{sat}} = 299.9$ K
 $h_f = 111.8$ kJ/kg, $h_{fg} = 2437.9$ kJ/kg
 $s_f = 0.391$ kJ/kg.K, $s_{fg} = 8.129$ kJ/kg.K
 $v_f = 0.001003$ m³/kg

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Q6. (b) A test on single cylinder, four stroke oil engine having bore diameter 18 cm and stroke 36 cm yielded the following results ;

Speed 285 rev/min ;

Brake torque 393 Nm ;

Indicated m.e.p. 7.2 bar ;

Fuel consumption 3.5 kg/hr

Cooling water flow 4.5 kg/min ;

Cooling water temperature rise 36°C ;

Air-fuel ratio by mass 25 ;

Exhaust gas temperature 415°C ;

Barometric pressure 1.013 bar ;

Room temperature 21°C ;

The fuel has a calorific value of 45200 kJ/kg and contains 15% by mass of hydrogen.

Determine :

- The indicated thermal efficiency.
- The volumetric efficiency based on atmospheric conditions.
- Draw up a heat balance in terms of kJ/min explaining clearly the content of such term.

Take $R = 0.287 \text{ kJ/kgK}$

C_v for dry exhaust gases = 1.005 kJ/kgK and for superheated steam $C_p = 2.05 \text{ kJ/kgK}$. **15**

Q6. (c) How does the ideal-gas refrigeration cycle differ from the Carnot refrigeration cycle ? **10**



- Q7. (a) Air compressed steadily by a reversible compressor from an inlet state of 100 kPa and 300 K to an exit pressure of 900 kPa. Determine the compressor work per unit mass for
- a) Isentropic compression with $K = 1.4$
 - b) Polytropic compression with $n = 1.3$
 - c) Isothermal compression and
 - d) Ideal two stage compression with intercooling with a polytropic exponent of 1.3. **15**
- Q7. (b) What is meant by delay period ? Tabulate the variables affecting delay period with their effects and reasons. **15**
- Q7. (c) Express the Bernoulli equation in three different ways using (a) energies (b) pressure and (c) heads. **10**
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- Q8. (a) Air enters the evaporator section of a window air conditioner at 100 kPa and 287°C with a volume flow rate of $6\text{ m}^3/\text{min}$. Refrigerant 134a at 120 kPa with quality of 0.3 enters the evaporator at a rate of 2 kg/min and leaves as saturated vapour at the same pressure. Determine the exit temperature of the air and the exergy destruction for this process assuming the outer surfaces of the air-conditioner are insulated.

Given properties

The gas constant of air is $0.287 \text{ kPa m}^3/\text{kgK}$

Specific heat of air at room temp. = 1.005 kJ/kgK

The properties of R – 134a at inlet and exit states are

$$P_1 = 120 \text{ kPa} \quad h_f = 22.47 \text{ kJ/kg} \quad h_{fg} = 214.52 \text{ kJ/kg}$$

$$s_f = 0.09269 \text{ kJ/kgK}, \quad s_{fg} = 0.85520 \text{ kJ/kgK}$$

$$h_g = 236.99 \text{ kJ/kg}$$

$$s_2 = 0.94789 \text{ kJ/kgK}$$

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Q8. (b) A centrifugal compressor has an impeller tip speed of 360 m/s. Determine :

- a) the absolute Mach number of flow leaving the radial vanes of the impeller and
- b) the mass flow rate

Given data :

- Impeller tip speed = 360 m/s
- Radial components of flow velocity at impeller exit = 30 m/s
- Slip factor = 0.9
- Flow area at impeller exit = 0.1 m^2
- Power input factor = 1.0
- Isentropic efficiency = 0.9
- Inlet stagnation temperature = 300 K
- Inlet stagnation pressure = 100 kN/m^2
- $R(\text{air}) = 287 \text{ J/kgK}$
- $\gamma(\text{air}) = 1.4$

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Q8. (c) Describe body forces and surface forces and explain how the net force acting on a control volume is determined. Is fluid weight a body force or surface force ? How about pressure ?

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